4a. Define $f: \mathbb{R}^2 \to \mathbb{R}$ by $f(0,0)=0$, $f(x,y)=\frac{xy^2}{x^2+y^4}$ for $(x,y) \neq (0,0)$. Prove that f is not continuous at $(0,0)$, but that restricted to any straight line through $(0,0)$ is continuous.

1) Let a sequence be defined as $\{(\frac{1}{n}, \frac{1}{\sqrt{n}}) \mid n \in \mathbb{N}\}$

The $\lim_{n\to\infty} (\frac{1}{n}, \frac{1}{\sqrt{n}}) = (0,0)$, (ie where $y=\sqrt{x}$)

but $f(x,y) = \frac{x \cdot y^2}{x^2+y^4} \Rightarrow f(x,\sqrt{x}) = \frac{x\cdot(\sqrt{x})^2}{x^2+(\sqrt{x})^2} = \frac{x^2}{x^2+x^2}$

so $\lim_{x\to 0} f(x,\sqrt{x}) = \lim_{x\to 0} \frac{x^2}{2x^2} = \frac{1}{2} \neq 0.$ ✓

2) If we restrict f to straight line approaches

then we have: i) $y=ax$ $a \in \mathbb{R}$

$$\lim_{x\to 0} f(x,ax) = \lim_{x\to 0} \frac{x\cdot a^2x^2}{x^2+a^4x^4} = \lim_{x\to 0} \frac{a^2x^3}{x^2+a^4x^4}$$

$$= \lim_{x\to 0} \frac{a^2x^3}{x^2+a^4x^4} \cdot \frac{1/x^3}{1/x^3} = \lim_{x\to 0} \frac{a^2}{\frac{1}{x}+a^4x^2} = 0$$

i) $x=0$

$$\lim_{y\to 0} (0,y) = \lim_{y\to 0} \frac{0\cdot y^2}{0^2+y^4} = \lim_{y\to 0} \frac{0}{y^4} = 0$$

$\Rightarrow$ so, if f is restricted to straight lines, then

$\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$, i.e. f is continuous at $(0,0)$ with the restriction.

4b. Prove that a polynomial f of odd degree has at least 1 real root.

From 3c we know f is continuous.

<u>Road Map</u>: will use IVT, so want to find a $x \in \mathbb{R}$ s.t. $f(x) > 0$ and a $y \in \mathbb{R}$ s.t. $f(y) < 0$.

let $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n+1} x^{2n+1}$

<u>W.L.O.G</u> suppose $a_{2n+1} > 0$

1st) if $a_{2n+1} x^{2n+1} > a_0 + a_1 x + a_2 x^2 + \dots + a_{2n+1} x^{2n}$ for some $x \in \mathbb{R}$ $(x > 0)$

then

$$f(x) > 0$$

$a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n} \leq |a_0| + |a_1 x| + |a_2 x^2| + \dots |a_{2n} x^{2n}|$

for $x > 1$, and since $x^{2n} \geq 0$

$< \sum_{i=0}^{2n} |a_i| x^{2n}$

so let's find and x s.t. $a_{2n+1} x^{2n+1} > x^{2n} \sum_{i=0}^{2n} |a_i|$

i.e. make $x > \dfrac{\sum_{i=0}^{2n} |a_i|}{a_{2n+1}}$ ← finite sum of finite values

$\Rightarrow$ If $x > \dfrac{\sum_{i=0}^{2n} |a_i|}{a_{2n+1}}$

$\Rightarrow f(x) > 0.$

2nd) if $|a_{2n+1} y^{2n+1}| > |a_0 + a_1 y + a_2 y^2 + \dots + a_{2n} y^{2n}|$

when $y^{2n+1} < 0$, then $f(y) < 0$

$|a_0 + a_1 y + a_2 y^2 + \dots + a_{2n} y^{2n}| \leq |a_0| + |a_1 y| + \dots + |a_{2n} y^{2n}|$

since $y^{2n} \geq 0$ and when $y < 1$

$|a_0| + |a_1 y| + \dots + |a_{2n} y^{2n}| < \sum_{i=0}^{2n} |a_i| y^{2n}$

so let $\sum_{i=0}^{2n} |a_i| y^{2n} < |a_{2n+1} y^{2n+1}|$

$\dfrac{\sum_{i=0}^{2n} |a_i|}{a_{2n+1}} < |y|$

since $y < 0$

i.e. $y < -\dfrac{\sum_{i=0}^{2n} |a_i|}{a_{2n+1}}$

$\Rightarrow f(y) < 0$

4b continued:

By ~~the~~ IVT, then $\exists\ z \in (y, x)$ s.t.

$f(z) = 0$ since $f(y) < 0 < f(x)$.

4c. Suppose (X,d) is a complete metric space. Let $f: X \to X$ be a map with $d(f(x), f(y)) \leq \frac{1}{2} d(x,y)$ for all $x, y \in X$. Show there exists a fixed point of f, i.e. a point $x \in X$ s.t. $f(x) = x$.

1st) We know $\exists r > 0$ s.t. $B_r(x_0) \Rightarrow f(B_r(x)) \subset B_r(x_0)$

(trivially this ball could be the whole space, since obviously $f(x) \subset X$).

<u>Claim</u> $B_r(x_0) \supsetneq f(B_r(x_0))$

$f(B_r(x_0)) \supsetneq f^2(B_r(x_0))$

$\vdots$

$f^n(B_r(x_0)) \supseteq f^{n-1}(B_r(x_0))$

and $\overline{f^n(B_r(x_0))} \supsetneq f^{n+1}(B_r(x_0))$.

<u>Proof</u>:

We are given $d(f(x), f(y)) \leq \frac{1}{2} d(x,y)$

Let's defined the diameter of a set B as the supremum $\{d(x,y) \mid x, y \in B\}$

So, for example, the diameter of $B_r(x_0)$ is $2r$ since $B_r(x_0) = (-r + x_0, x_0 + r)$ i.e. all the points in the interval.

Let diameter $f^n(B_r(x_0))$ be c.

Since the supremum $\{d(x,y) \mid x, y \in f^n(B_r(x_0))\}$ = diameter of $f^n(B_r(x_0))$.

and $d(f(x), f(y)) \leq \frac{1}{2} d(x,y)$

then $\forall f^{n+1}(x), f^{n+1}(y) \in f^{n+1}(B_r(x_0))$ $(x, y \in B_r(x_0))$

are s.t. $d(f^{n+1}(x), f^{n+1}(y)) \leq \frac{1}{2} \cdot$ diameter $f^n(B_r(x_0)) = \frac{c}{2}$

This process leads us to arguing $f^n(B_r(x_0)) \supsetneq f^{n+1}(B_r(x_0))$ ✓

<u>Now</u>, to show $\overline{f^n(B_r(x_0))} \supsetneq \overline{f^{n+1}(B_r(x_0))}$

Since $f^n(B_r(x_0))$ contains points apport $> \frac{1}{2}$ diameter $f^n(B_r(x_0))$ and $f^{n+1}(B_r(x_0))$ only contains points apport $\leq \frac{1}{2}$ diameter $f^n(B_r(x_0))$

$\overline{f^n(B_r(x_0))} \supsetneq \overline{f^{n+1}(B_r(x_0))}$ ✓

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4c. continued.

So, now that our claim is proven,
i.e. $\overline{f^n(B_r(x_0))} \supsetneq \overline{f^{n+1}(B_r(x_0))}$ $\forall n \geq 0$

Then, by the cantor intersection lemma

$\bigcap_{n=0}^{\infty} \overline{f^n(B_r(x_0))}$ is nonempty, let $\bigcap_{n=0}^{\infty} f^n(B_r(x_0)) = p$.

so, since $f^n(p) \in \bigcap_{n=0}^{\infty} \overline{f^n(B_r(x_0))}$ (trivially true)

from these, we can conclude

$$f^n(p) = p$$

[W.T.S. $f(p) = p$]

<u>mini proof by contraction:</u>

Suppose $f(p) = z$, and $z \neq p$

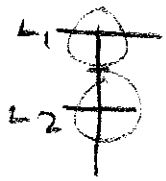
then, since $f^{n+1}(p) \in \bigcap_{x=0}^{\infty} f^n(B_r(x_0))$

$$p = f^{n+1}(p) = f \circ f^n(p) = f(p) = z$$

$\Rightarrow p = z$, contradicts, $z \neq p$.

so $f(p) = p$, where $p \in \bigcap_{n=0}^{\infty} \overline{f^n(B_r(x_0))}$

4d. Prove that if $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} g(x) = L_2$, and $f(x) \le g(x)$ for x in some interval containing a, then $L_1 \le L_2$.



<u>Proof</u> (by contradiction)

Suppose not, i.e. $L_1 > L_2$

(by density of $\mathbb{Q}$) $\exists r \in \mathbb{Q}$ s.t. $L_1 < r < L_2$

more specifically $L_2 < L_2 + \frac{L_1 - L_2}{2} < L$

Let $\varepsilon = \frac{L_1 - L_2}{2}$, so $B_\varepsilon(L_1) \cap B_\varepsilon(L_2) = \emptyset$.

From definition of (i) $\lim_{x\to a} f(x) = L_1$ and (ii) $\lim_{x\to a} g(x) = L_2$

(i) $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |f(x) - L_1| < \varepsilon$

(ii) $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |g(x) - L_2| < \varepsilon$

Let the ε be specified as $\varepsilon = \frac{L_2 - L_1}{2}$

then $\exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |f(x) - L_1| < \varepsilon$

and $\Rightarrow |g(x) - L_2| < \varepsilon$

$\Rightarrow$ $f(x) \in (-\varepsilon + L_1, L_1 + \varepsilon)$

$g(x) \in (-\varepsilon + L_2, L_2 + \varepsilon)$

i.e. $f(x) < L_1 + \varepsilon = L_1 + \frac{L_1 - L_2}{2} = \frac{L_1 + L_2}{2}$

and $g(x) > L_2 - \varepsilon = \frac{L_1 + L_2}{2}$

So $f(x) < L_1 + \varepsilon = L_2 - \varepsilon < g(x)$

$\Rightarrow f(x) < g(x) \quad \forall x \in B_\delta(a)$

but we know $\exists \gamma > 0$ s.t.

$\forall x \in B_\gamma(a) \Rightarrow f(x) < g(x)$

so $\exists x \in B_\delta(a)$ s.t. $f(x) \not< g(x)$

so we have a contradiction $\Rightarrow L_1 \le L_2$.

4e. Prove the squeeze lemma. Suppose $f(x) \le g(x) \le h(x)$ for some open interval around a. Suppose $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$. Then, $\lim_{x\to a} g(x)$ exists and is L.

(Make sure you prove that the limit actually exists.)

From the definition of $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} h(x) = L$, we have:

$\forall \varepsilon > 0 \ \exists \delta_1 > 0$ s.t.
$|x-a| < \delta_1 \Rightarrow |f(x) - L| < \varepsilon$
$\Downarrow$
$f(x) \in (L-\varepsilon, L+\varepsilon)$

$\forall \varepsilon > 0 \ \exists \delta_2 > 0$ s.t. $|x-a| < \delta_2$
$\Rightarrow |h(x) - L| < \varepsilon$
$\Downarrow$
$h(x) \in (L-\varepsilon, L+\varepsilon)$

$L - \varepsilon < f(x)$ and $h(x) < L + \varepsilon$

for $x \in B_\gamma(a)$ $f(x) \le g(x) \le h(x)$,

so for $\delta = \min(\gamma, \delta_1, \delta_2)$

$L - \varepsilon < f(x) \le g(x) \le h(x) < L + \varepsilon$

is true $\Rightarrow$ $\forall \varepsilon > 0, \exists \delta > 0$ s.t. (specifically)

s.t. $|x-a| < \delta \Rightarrow |g(x) - L| < \varepsilon$

so $\lim_{x\to a} (g(x)) = L$.